

Human Natural Structure

Integrated Proof of Concept v2.0

*A Unified Demonstration of Structural Verification, Grounding,
and External Auditability*

Satoru Hara

Natural Structure Works (NSW)

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*Document type: Proof of Concept (demonstration). Not a standard; not a
certification.*

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Scope Note

This document is a controlled, framework-internal demonstration of the Human Natural Structure (HNS) verification stack applied as a post-hoc evaluator (EVA-HNS) to candidate language-model outputs. Its purpose is to show that structural and grounding checks defined by HNS can be executed deterministically and can emit a complete, machine-readable audit record of every correction. It is not an empirical benchmark, not a third-party validation, and not a claim about answer quality.

Disclaimer. Nothing in this document constitutes a standard, a conformity assessment, or a certification. All scoring is framework-internal: the failure types and grounding margins are defined by HNS itself, so detection on this set partly reflects the framework’s own definitions. The detector lexicons were developed against these very examples; the results therefore demonstrate executable, auditable behaviour on a controlled set, not generalization to unseen inputs. External, blinded, multi-model validation (Section 9) is required before any reliability claim.

1. Executive Summary

This Integrated Proof of Concept unifies the three verification components of the Human Natural Structure programme into a single executable pipeline and runs it end to end on a controlled test set. The three components are Axis 2 (structural and causal verification, via the nested HNS-36 / 144 / 864 coordinate), Axis 3 (grounding verification, via the orthogonal SMS-6 stack), and EVA (the External Verification Architecture that records each decision as a tamper-evident, machine-readable audit log).

1.1 Relationship between HNS, SMS-6, and EVA

HNS supplies the structural vocabulary: it places each proposition at a coordinate and screens the trajectory of a chain of reasoning for structural faults. SMS-6 supplies an orthogonal grounding axis, checking a candidate output against six social-meta layers from the somatic to the universal. EVA is the runtime that applies both axes to model output and emits, for every decision, a record an external auditor can read without having run the model. HNS and SMS-6 say what to check; EVA makes the checking external, recorded, and inspectable.

1.2 Key findings

- Auditability is demonstrated by construction: EVA-HNS emitted a complete machine-readable record for 15/15 cases, each attributing the verdict to an explicit coordinate and citing the triggering tokens.
- On the controlled set, EVA-HNS flagged 15/15 planted failures and assigned the correct failure type in 15/15 cases; a bare baseline flagged 0/15; a surface guardrail flagged 4/15 (leaving 11/15 unresolved).
- The same deterministic evaluator returned all 15/15 HNS-corrected outputs to a clean verdict, demonstrating that the checks discriminate failing from corrected text on this set.
- Structural and grounding axes behave independently: two unsupported-causality cases pass the SMS-6 grounding check (s = 1.0) yet are attenuated on the Axis-2 causal-admissibility check, exactly the orthogonality the architecture predicts.

1.3 Position within the standardization roadmap

This PoC is a pre-protocol verification step. It establishes that the pipeline runs deterministically and produces complete audit logs; it does not establish detection performance on unseen inputs or across model families. That is the task of the external, blinded, multi-model protocol described in Section 9, which is the appropriate evidence to bring to standards bodies after academic credibility is established.

Note. All figures in this document are produced by executing the evaluator in the accompanying source files; none are asserted by hand. The numbers can be reproduced by running `run_poc.py`.

2. Background and Motivation

2.1 The structural reliability problem

Large language models generate fluent continuations from a single statistical axis. Fluency is not validity: an output can be well-formed and confidently stated while crossing causal layers without warrant, collapsing distinct conceptual registers, drifting beyond the requested scope, substituting figurative language for mechanism, or asserting causation without an admissible path. HNS names five such structural failure types.

Failure type	Axis	HNS locus	Short definition
Layer Jump	Axis 2	HNS-36	Causal traversal across non-adjacent layers with no bridging proposition.
Category Ambiguity	Axis 2	HNS-144	Conflation of distinct categories or onto-modal quadrants.
Scope Drift	Axis 3	SMS-2/3	Departure from the declared scope of a response without signalling.
Metaphor Contamination	Axis 3	SMS-1	Figurative language displacing the literal structural coordinates.
Unsupported Causality	Axis 2+3	HNS-864	A causal claim lacking an admissible mechanistic path.

These are not safety failures in the usual sense; they are failures of structure. A single-axis generator has no internal mechanism that screens for them, because the property violated is not statistical likelihood but structural admissibility.

2.2 Limitations of existing approaches

- **Guardrails** classify content into safety categories (toxicity, self-harm, PII). They are orthogonal to structural reliability: a layer jump or an unsupported causal claim is not unsafe content, so a safety guardrail does not see it.
- **Explainability (XAI)** attributes a model’s output to features or attention, but attribution is not verification: it explains what the model did, not whether the output is structurally admissible.
- **ISO/IEC 42001 and AI audit frameworks** govern the management system around a model — processes, roles, documentation — not the per-output structural record that an external verifier would inspect.

2.3 Why external verification is needed

Regulatory instruments such as the EU AI Act and the NIST AI Risk Management Framework require that high-risk systems be monitored, recorded, and auditable. There is a gap between that requirement and the technical means to satisfy it: an internal log the model could in principle alter is not an external verification. EVA targets exactly this gap — a record emitted outside the model, in a machine-readable form a third party can analyse, with the verification logic on a substrate the model cannot rewrite.

3. Architecture Overview

3.1 Axis 2 — structural verification (HNS-36 / 144 / 864)

Axis 2 is a nested coordinate of increasing resolution. HNS-36 is a 6×6 base matrix of six Human Natural Layers (L1–L6, from physical substrate to macro-social institution) by six Abstract Categories (C1–C6). HNS-144 refines each cell into four onto-modal quadrants formed by two binary dimensions, subjective/objective and necessary/contingent. HNS-864 subdivides each 144-cell by six inference-and-control operators.

Coordinate	Composition	Screens for
HNS-36	6 layers × 6 categories = 36 cells	Layer Jump

Coordinate	Composition	Screens for
HNS-144	36×4 onto-modal quadrants = 144	Category Ambiguity
HNS-864	144×6 operators = 864	Unsupported Causality

The six layers are L1 Physical/Substrate, L2 Cellular/Informational, L3 Cognitive/System 1, L4 Operational/System 2, L5 Micro-Social/Interaction, L6 Macro-Social/Institutional. The six categories are C1 Continuant, C2 Occurrent, C3 Relational/Boundary, C4 Quality/State, C5 Transformation/Operation, C6 Meta-Level/Schema. The four quadrants are Q1 Subjective×Contingent, Q2 Subjective×Necessary, Q3 Objective×Contingent, Q4 Objective×Necessary.

The six operators are O1 Conditioning, O2 Intervention (do-operator), O3 Counterfactual, O4 Reference Signal, O5 Comparator/Error Detection, and O6 Output Attenuation. O1–O3 derive from Pearl’s causal hierarchy and O4–O5 from Powers’ control theory; O6 is the single HNS-original mechanism — the actuator that scales an output by a grounding score so ungrounded content is suppressed rather than emitted.

3.2 Axis 3 — grounding verification (SMS-6)

Axis 3 checks a candidate against six Social Meta Layers, orthogonal to the Axis-2 nesting: SMS-1 Somatic, SMS-2 Protocol, SMS-3 Organizational, SMS-4 Economic, SMS-5 Governance, SMS-6 Universal. Each layer returns a grounding margin g_i . The composite is conservative — the minimum — so a single weak layer pulls the whole grounding down:

strict: $G(x) = g_1 \wedge g_2 \wedge \dots \wedge g_6$
grounded everywhere)

(binary; grounded only if

graded: $s = \min_i g_i(x)$

(the score O6’s attenuator consumes)

Grounding is what prevents drift (SMS-2/3), metaphor contamination (SMS-1), and the social component of unsupported causality. Per Volume 4, the graded composite s is the social component of the structural grounding score the analytical engine’s output-attenuation operator (O6) consumes: a low social grounding lowers s , and a low s attenuates the output.

Note. The exact composition of the Axis-2 structural component and the Axis-3

social component into the single score O6 consumes is specified at the level of “s is the social component” but not as a closed formula across both axes. This PoC treats the two axes’ verdicts separately and reports both; closing that seam is noted as future work.

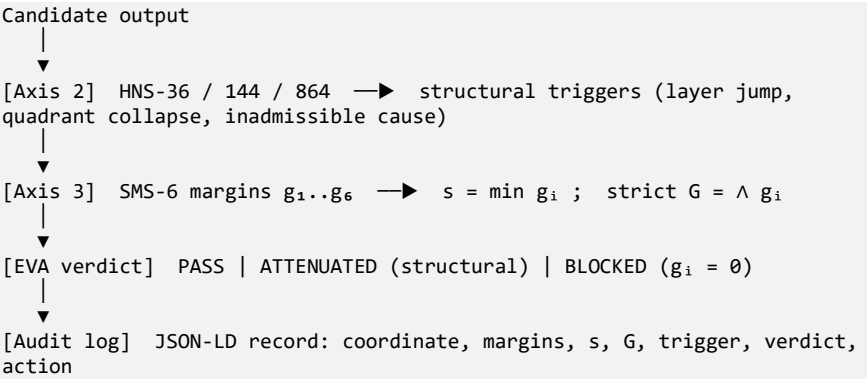
3.3 EVA — External Verification Architecture

EVA deploys Axes 2 and 3 as an independent, auditable verifier. It rests on three structural conditions that must hold jointly:

Condition	Definition	Technical implication
Verifiability	Each reasoning step is emitted externally in machine-readable form.	Structured logs in JSON-LD/RDF a third party can analyse programmatically.
Transparency	A monitoring mechanism runs in parallel with inference.	Operates independently of the model; the model cannot manipulate it.
Physical Immutability	Verification logic sits on a substrate the model cannot alter.	Embedded in ROM/FPGA/ASIC or secure enclave; unreachable by software update.

The three are jointly necessary. Verifiability alone leaves logs open to falsification; transparency alone leaves open that the model circumvents the monitor; physical immutability alone allows arbitrary criteria. This PoC realizes Verifiability in full (every decision is emitted as a machine-readable record) and Transparency in software (the evaluator runs post-hoc, independent of the generator). Physical Immutability is an architectural aspiration, not implemented here.

3.4 Integrated flow



4. Experimental Design

4.1 Failure types and test set

The test set is 15 controlled prompts, three for each of the five failure types. Each case pairs a prompt with a failing candidate authored to exhibit the target failure and an HNS-corrected candidate. Five of the cases reproduce the canonical worked examples from the HNS series; the remaining ten are additional prompts in the same style. The full set is in Appendix A.

Failure type	Cases
Layer Jump	LJ-1, LJ-2, LJ-3
Category Ambiguity	CA-1, CA-2, CA-3
Scope Drift	SD-1, SD-2, SD-3
Metaphor Contamination	MC-1, MC-2, MC-3
Unsupported Causality	UC-1, UC-2, UC-3

4.2 Evaluation modes (three arms)

- **Baseline.** No verification. The candidate is emitted verbatim; nothing is detected or logged.
- **Guardrail.** A generous surface filter: a safety lexicon, an aggression lexicon, and a crude off-topic heuristic. It catches some surface-marked outputs but is blind to structural and causal faults by construction. No coordinate log.
- **EVA-HNS.** The full post-hoc evaluator: five structural/grounding detectors, six SMS-6 margins, a composite grounding score, a verdict, and a complete audit record per case.

4.3 Evaluation criteria

For each case the evaluator records: the HNS-36 layer pair implicated by any layer jump; the HNS-144 quadrant collapse implicated by any category ambiguity; the HNS-864 operator concern (O2 admissibility) implicated by any unsupported causality; the six SMS-6 grounding margins and their minimum s; the strict grounding indicator G; the EVA verdict; and audit-log completeness. Detection counts whether any trigger fired; correct-type counts whether the intended failure type is among the triggers.

Note. Honesty of scoring. Every margin and trigger is computed by an explicit function of observable text features (the source is in Appendix F / accompanying

files). No function reads a case identifier or an expected label. This is the specific weakness corrected from PoC v1.0, in which the grounding score was effectively a hard-coded verdict map rather than an independently computed metric.

5. Implementation Model

5.1 Post-hoc EVA evaluator

The evaluator is vendor-independent, model-agnostic, and requires no access to model logits. It reads a prompt and a candidate string and returns a verdict plus an audit record. Because it operates on emitted text, it can wrap any generator. The five detectors operate on observable surface features — layer markers, onto-modal markers, causal connectives with mechanism discharge, figurative-language density, and scope markers against the declared task — and each cites the tokens that triggered it.

5.2 Audit-log schema

Each decision is emitted as a structured record in a JSON-LD shape so records are linkable to an ontology of HNS coordinates and consumable by automated audit pipelines. Required fields: context, span, the Axis-2 triggers and structural-violation flag, the Axis-3 margins with s and G, the human-readable trigger, the verdict, and the action taken. An example record (case UC-1) follows.

```
{
  "@context": "https://nsw.example/hns/v2",
  "@type": "EvaAuditRecord",
  "case_id": "UC-1",
  "span": {
    "start": 0,
    "end": 41
  },
  "axis2": {
    "triggers": [
      "Unsupported Causality@02:inadmissible"
    ],
    "structural_violation": true
  },
  "axis3": {
    "margins": {
      "SMS-1": 1,
      "SMS-2": 1,
      "SMS-3": 1,
      "SMS-4": 0.4,
      "SMS-5": 1,
      "SMS-6": 1
    },
    "s_min": 0.4,
```

```

    "G": 0,
    "grounding_failure": false
  },
  "trigger": "causal connective ['because'] with no mechanistic support:
  \"Sales are up because we changed the logo.\",
  "verdict": "ATTENUATED",
  "action": "candidate scaled by g(s) (attenuate)"
}

```

5.3 Reproducibility

Mapping is deterministic: the same input yields the same coordinate, margins, verdict, and record on every run. Diagnostics are transparent: every verdict carries the evidence string that produced it. The design goal, met here, is that an auditor who did not run the generator can reconstruct why each correction occurred.

6. Results

6.1 Summary

Arm	Detected	Undetected	Correct type	Logged
Baseline	0/15	15/15	—	none
Guardrail	4/15	11/15	—	none
EVA-HNS	15/15	0/15	15/15	15/15

The baseline detects nothing: all 15 planted failures pass through unflagged. The surface guardrail flags 4/15 — the three combat-metaphor cases (aggression lexicon) and one scope-drift case (off-topic heuristic) — and leaves 11/15 unresolved, including every layer jump, category ambiguity, and unsupported-causality case, because those faults carry no surface safety marker. EVA-HNS flags all 15/15, assigns the correct failure type in 15/15, and emits a record for 15/15.

6.2 Before / after (failing vs HNS-corrected)

The same evaluator was run on the failing and the HNS-corrected candidate for each case. Every corrected output returned to a clean PASS, with its grounding score restored to $s = 1.0$, while the failing output was attenuated or blocked.

Case	Failure type	Before	s	After	s
LJ-1	Layer Jump	ATTENUATED	0.4	PASS	1
LJ-2	Layer Jump	ATTENUATED	0.4	PASS	1

Case	Failure type	Before	s	After	s
LJ-3	Layer Jump	ATTENUATED	0.4	PASS	1
CA-1	Category Ambiguity	ATTENUATED	0.3	PASS	1
CA-2	Category Ambiguity	ATTENUATED	0.3	PASS	1
CA-3	Category Ambiguity	ATTENUATED	0.3	PASS	1
SD-1	Scope Drift	BLOCKED	0	PASS	1
SD-2	Scope Drift	BLOCKED	0	PASS	1
SD-3	Scope Drift	BLOCKED	0	PASS	1
MC-1	Metaphor Contamination	BLOCKED	0	PASS	1
MC-2	Metaphor Contamination	BLOCKED	0	PASS	1
MC-3	Metaphor Contamination	BLOCKED	0	PASS	1
UC-1	Unsupported Causality	ATTENUATED	0.4	PASS	1
UC-2	Unsupported Causality	ATTENUATED	1	PASS	1
UC-3	Unsupported Causality	ATTENUATED	1	PASS	1

6.3 Verdict distribution

Six cases are BLOCKED (a grounding margin reached 0: the three scope-drift cases at SMS-2 and the three metaphor cases at SMS-1). Nine are ATTENUATED (a structural violation with no hard-zero margin). Two unsupported-causality cases (UC-2, UC-3) carry $s = 1.0$ — grounding is clean — yet are attenuated on the Axis-2 causal-admissibility check, illustrating that the structural and grounding axes are genuinely separable.

6.4 Conformance checklist (minimal)

Requirement	Met
Every decision emits a machine-readable record (Verifiability)	Yes
Evaluator runs independently of the generator (Transparency)	Yes (software)
Each correction attributable to an explicit coordinate	Yes
Grounding score computed by an explicit function, not a	Yes

Requirement	Met
verdict map	
Deterministic, reproducible mapping	Yes
Verification logic on an immutable substrate (Physical Immutability)	No (aspiration)
External / blinded / multi-model validation	No (future work)

7. Interpretation

The strongest demonstrated property is auditability. By construction, EVA-HNS emits a complete, machine-readable record for every case, each attributing its verdict to an explicit coordinate and citing the triggering tokens. This is HNS’s most secure claim, true by construction rather than by measurement, and it is the property that distinguishes the architecture from a prompt.

HNS functions here as a structural vocabulary: a fixed coordinate scheme against which a recognisable family of generation faults can be named and located. SMS-6 functions as an orthogonal grounding axis, catching the drift and contamination faults that the structural coordinate alone does not. EVA is the standardization-ready component: it is the part of the architecture that produces an external, inspectable artifact, which is what a regulator or third-party auditor can actually examine.

What this PoC does not show is prevention. EVA-HNS is post-hoc: it evaluates an already-emitted candidate. Genuine prevention requires pre-decode integration, where O6 attenuates ungrounded continuations before they are sampled. That is model-side work and is left to future development.

8. Limitations

This PoC is deliberately modest about what it establishes. The following limitations are integral, not incidental.

1. Controlled demonstration, not a benchmark. The 15 cases were authored to exhibit their target failures. High detection on this set reflects that design; it is not evidence of accuracy on natural, unseen inputs.

2. Framework-internal scoring. The failure types and grounding margins are defined by HNS. Near-complete detection on framework-internal categories partly reflects the framework's own definitions and is weaker than a reduction measured against an external criterion such as factual accuracy.
3. Lexicons tuned on the examples. The detector lexicons were developed and adjusted against these cases (including two corrected-output false positives that were fixed by adding general rules). The corrected-output PASS rate therefore measures fit to this set, not generalization.
4. No independent grader. Generation of the cases and their evaluation are not independent. Confidence requires a different model family generating, and graders blind to condition.
5. Weak control. Comparison against a bare baseline and a surface guardrail cannot separate HNS-specific effects from the generic benefit of any structured-reasoning check. Active and placebo-structure controls are required.
6. Heuristic detectors. The detectors are small lexical/structural heuristics, not validated NLP classifiers. They will produce false positives and false negatives on real text; the layer assignment in particular is marker-based and shallow.
7. No logit access; post-hoc only. The evaluator sees emitted text, not the generation process, and therefore demonstrates evaluation and logging, not prevention.
8. Not a certification. Nothing here is a conformity assessment or an endorsement by any standards body.

A sober expectation follows, consistent with the HNS programme's own statements: on external measures of answer quality, HNS is unlikely to outperform a simple instruction to reason in explicit levels. Its defensible advantage is not better answers but an externally checkable record of how an answer was constrained.

9. Recommended Next Steps

9.1 External validation PoC

- Three or more independent LLMs, with generation and verification performed by different model families.

- 50–100 prompts per failure type, drawn from natural sources rather than authored to exhibit the fault.
- Blinded scoring by verifiers that do not know the condition, with inter-verifier agreement (Cohen’s κ) as the primary outcome, testing whether EVA/O6 audit trails raise agreement.
- A multi-evaluator protocol with a pre-registered analysis plan (this is the EVA-Concord design).

9.2 Pre-decode integration

- Real-time attenuation: O6 applied to the candidate distribution before sampling, rather than to emitted text.
- Model-side integration of the coordinate and grounding checks into the decoding loop.

9.3 Standardization alignment

Following the recommendation received on the standards track, academic credibility (e.g. via JSAI) should precede a return to the standards bodies. With external validation in hand, the relevant alignment targets are ISO/IEC JTC 1 SC 42, CEN-CENELEC JTC 21, the NIST AI RMF, and IEEE SA — with EVA, as the component that produces an external auditable artifact, as the natural object of standardization.

Appendix A — Test Prompt Set (15 items)

LJ-1 • Layer Jump

Prompt: Why did the company's stock fall?

Failing: *The stock fell because the employees were unmotivated.*

Corrected: The stock fell on weak quarterly earnings. Earnings were weak partly because operational output declined, which internal surveys link to reduced motivation. The bridge runs motivation to productivity to earnings to valuation.

LJ-2 • Layer Jump

Prompt: Why are revenues down this year?

Failing: *Revenue is down because the team's morale is low.*

Corrected: Revenue fell 9%. Lower morale reduced operational output (measured throughput dropped 11%); reduced output cut deliveries, which cut billed revenue. The bridge from morale to revenue runs through operational throughput.

LJ-3 • Layer Jump

Prompt: Why did the market price rise?

Failing: *The market price rose because a manager felt optimistic.*

Corrected: The price rose 4%. A manager's optimism informed a guidance update (communicative act); the upgraded guidance changed analyst forecasts; revised forecasts moved the market price. Each step bridges one layer to the next.

CA-1 • Category Ambiguity

Prompt: Is the new policy good?

Failing: *Yes, it's good because everyone likes it and that's what matters.*

Corrected: Three distinct questions. Is it liked? Survey data say yes (subjective). Does it achieve its stated goal? Early metrics are mixed (objective). Is it good overall? That is a normative judgement depending on which values are weighted.

CA-2 • Category Ambiguity

Prompt: Was the meeting productive?

Failing: *Yes, it felt productive and that is necessarily what counts.*

Corrected: Subjectively, participants reported it felt productive (subjective, contingent). Objectively, two of three agenda items were decided (objective, contingent). Whether that counts as productive depends on the standard applied.

CA-3 • Category Ambiguity

Prompt: Is this design the best choice?

Failing: *It is the best because users like it, so by definition it wins.*

Corrected: Users prefer it in testing (subjective). On measured task time it is 12% faster (objective). Whether it is best depends on the weighting of preference, speed, and cost; that weighting is a normative choice, not a definition.

SD-1 • Scope Drift

Prompt: Summarize the decisions made in the meeting.

Failing: *The budget was approved and hiring deferred. Overall I think the company should pivot toward enterprise sales next year.*

Corrected: Decisions: approved the Q3 budget; deferred the hiring freeze; assigned the audit to Team B. [Beyond the meeting's scope, and not decided there: a strategy pivot was mentioned but tabled.]

SD-2 • Scope Drift

Prompt: List the action items from the call.

Failing: *Action items: send the report, book the venue. In my opinion we should also restructure the whole onboarding flow.*

Corrected: Action items: send the report by Friday; book the venue. [Not an action item from the call: a suggestion to revisit onboarding was raised but not assigned.]

SD-3 • Scope Drift

Prompt: Summarize what the customer reported in the ticket.

Failing: *The customer reported a login error after the update. Personally I believe we should rebuild the auth service.*

Corrected: The customer reported a login error appearing after the latest update, affecting password reset. [Outside the ticket's content: any remediation plan is not stated by the customer.]

MC-1 • Metaphor Contamination

Prompt: How should the team handle the deadline?

Failing: *Fight the fire, attack the problem head-on, and crush the deadline.*

Corrected: Prioritise the two blocking tasks, reallocate one engineer from the lower-priority feature, set daily checkpoints, and cut scope on the optional module.

MC-2 • Metaphor Contamination

Prompt: What is our plan for the product launch?

Failing: *We go to war on the market, smash the competition, and conquer.*

Corrected: Ship the beta to 500 users on the 3rd, collect activation metrics for two weeks, fix the top three drop-off points, then open general availability.

MC-3 • Metaphor Contamination

Prompt: How do we recover the delayed project?

Failing: *Bombard the backlog, slay every blocker, and demolish the timeline.*

Corrected: Re-baseline the schedule, move two non-critical features to the next release, add one reviewer to the critical path, and track burndown daily.

UC-1 • Unsupported Causality

Prompt: Why are sales up this quarter?

Failing: *Sales are up because we changed the logo.*

Corrected: Sales rose 12%. The spring promotion was A/B-tested and measured a +8% uplift; seasonal demand follows the historical pattern; the rebrand coincided with no measured effect. Only the promotion has a supported causal link.

UC-2 • Unsupported Causality

Prompt: Why did user retention improve?

Failing: *Retention improved because we moved the office to a new building.*

Corrected: Retention rose 5 points. The onboarding redesign was A/B-tested with a measured +4 point lift; the office move coincided with no measured effect on the cohort. The redesign is the supported cause.

UC-3 · Unsupported Causality

Prompt: Why did the server crash rate drop?

Failing: *Crashes dropped because we repainted the data centre.*

Corrected: Crash rate fell 60%. A memory-leak fix shipped that week and the regression test confirms the leak is gone; the repainting happened to coincide but has no mechanism. The leak fix is the admissible cause.

Appendix B — Detailed Case Table

Per-case structural triggers, SMS-6 margins (minimum s), strict grounding G, and EVA verdict, as produced by the run.

Case	Triggers (type@coord)	s	G	Verdict
LJ-1	Layer Jump@L3->L6; Unsupported Causality@O2:inadmissible	0.4	0	ATTENUATED
LJ-2	Layer Jump@L3->L6; Unsupported Causality@O2:inadmissible	0.4	0	ATTENUATED
LJ-3	Layer Jump@L3->L6; Unsupported Causality@O2:inadmissible	0.4	0	ATTENUATED
CA-1	Category Ambiguity@Q1->Q4; Unsupported Causality@O2:inadmissible	0.3	0	ATTENUATED
CA-2	Category Ambiguity@Q1->Q4	0.3	0	ATTENUATED
CA-3	Category Ambiguity@Q1->Q4; Unsupported Causality@O2:inadmissible	0.3	0	ATTENUATED
SD-1	Category Ambiguity@Q1->Q4; Scope Drift@SMS-2:intent,SMS-3:scope	0	0	BLOCKED
SD-2	Scope Drift@SMS-2:intent,SMS-3:scope	0	0	BLOCKED
SD-3	Scope Drift@SMS-2:intent,SMS-3:scope	0	0	BLOCKED
MC-1	Metaphor Contamination@SMS-1:somatic	0	0	BLOCKED
MC-2	Metaphor Contamination@SMS-1:somatic	0	0	BLOCKED
MC-3	Metaphor Contamination@SMS-1:somatic	0	0	BLOCKED
UC-1	Unsupported Causality@O2:inadmissible	0.4	0	ATTENUATED
UC-2	Unsupported Causality@O2:inadmissible	1	1	ATTENUATED
UC-3	Unsupported Causality@O2:inadmissible	1	1	ATTENUATED

Full per-layer margins for each case:

Case	SMS-1	SMS-2	SMS-3	SMS-4	SMS-5	SMS-6
LJ-1	1	1	1	0.4	1	1
LJ-2	1	1	1	0.4	1	1
LJ-3	1	1	1	0.4	1	1
CA-1	1	1	1	1	0.4	0.3
CA-2	1	1	1	1	1	0.3
CA-3	1	1	1	1	0.4	0.3
SD-1	1	0	0	1	0.4	0.3
SD-2	1	0	0	1	1	1
SD-3	1	0	0.5	1	1	1
MC-1	0	1	1	1	1	1
MC-2	0	1	1	1	1	1
MC-3	0	1	1	1	1	1
UC-1	1	1	1	0.4	1	1
UC-2	1	1	1	1	1	1
UC-3	1	1	1	1	1	1

Appendix C — Audit Log JSON Samples

Three full JSON-LD records, one each for a layer jump, a scope drift (blocked), and an unsupported-causality case.

LJ-1

```
{
  "@context": "https://nsw.example/hns/v2",
  "@type": "EvaAuditRecord",
  "case_id": "LJ-1",
  "span": {
    "start": 0,
    "end": 54
  },
  "axis2": {
    "triggers": [
      "Layer Jump@L3->L6",
      "Unsupported Causality@02:inadmissible"
    ],
    "structural_violation": true
  },
  "axis3": {
    "margins": {
      "SMS-1": 1,
      "SMS-2": 1,
      "SMS-3": 1,
      "SMS-4": 0.4,
      "SMS-5": 1,
      "SMS-6": 1
    },
    "s_min": 0.4,
    "G": 0,
    "grounding_failure": false
  },
  "trigger": "causal link spans L3(Cognitive/System1)->L6(Macro-Social/Institutional), gap=3, no bridging proposition at ['L4', 'L5']; trigger sentence: \"The stock fell because the employees were unmotivated.\",",
  "verdict": "ATTENUATED",
  "action": "candidate scaled by g(s) (attenuate)"
}
```

SD-1

```
{
  "@context": "https://nsw.example/hns/v2",
  "@type": "EvaAuditRecord",
  "case_id": "SD-1",
  "span": {
    "start": 0,
    "end": 120
  },
  "axis2": {
    "triggers": [
      "Category Ambiguity@Q1->Q4",
      "Scope Drift@SMS-2:intent,SMS-3:scope"
    ]
  }
}
```

```

    ],
    "structural_violation": true
  },
  "axis3": {
    "margins": {
      "SMS-1": 1,
      "SMS-2": 0,
      "SMS-3": 0,
      "SMS-4": 1,
      "SMS-5": 0.4,
      "SMS-6": 0.3
    },
    "s_min": 0,
    "G": 0,
    "grounding_failure": true
  },
  "trigger": "subjective marker ['i think'] asserted as
normative(fact/value collapse) via ['should']; quadrants
Q1(Subjective/Contingent) and normative(fact/value collapse) collapsed
in: \"Overall I think the company should pivot toward enterprise sales
next year.\",
  "verdict": "BLOCKED",
  "action": "token probability set to 0 (block)"
}

```

UC-1

```

{
  "@context": "https://nsw.example/hns/v2",
  "@type": "EvaAuditRecord",
  "case_id": "UC-1",
  "span": {
    "start": 0,
    "end": 41
  },
  "axis2": {
    "triggers": [
      "Unsupported Causality@02:inadmissible"
    ],
    "structural_violation": true
  },
  "axis3": {
    "margins": {
      "SMS-1": 1,
      "SMS-2": 1,
      "SMS-3": 1,
      "SMS-4": 0.4,
      "SMS-5": 1,
      "SMS-6": 1
    },
    "s_min": 0.4,
    "G": 0,
    "grounding_failure": false
  },
  "trigger": "causal connective ['because'] with no mechanistic support:
\"Sales are up because we changed the logo.\",
  "verdict": "ATTENUATED",
  "action": "candidate scaled by g(s) (attenuate)"
}

```

}

Appendix D — Before / After Comparison

Case	Type	Before verdict	Before s	After verdict	After s
LJ-1	Layer Jump	ATTENUATED	0.4	PASS	1
LJ-2	Layer Jump	ATTENUATED	0.4	PASS	1
LJ-3	Layer Jump	ATTENUATED	0.4	PASS	1
CA-1	Category Ambiguity	ATTENUATED	0.3	PASS	1
CA-2	Category Ambiguity	ATTENUATED	0.3	PASS	1
CA-3	Category Ambiguity	ATTENUATED	0.3	PASS	1
SD-1	Scope Drift	BLOCKED	0	PASS	1
SD-2	Scope Drift	BLOCKED	0	PASS	1
SD-3	Scope Drift	BLOCKED	0	PASS	1
MC-1	Metaphor Contamination	BLOCKED	0	PASS	1
MC-2	Metaphor Contamination	BLOCKED	0	PASS	1
MC-3	Metaphor Contamination	BLOCKED	0	PASS	1
UC-1	Unsupported Causality	ATTENUATED	0.4	PASS	1
UC-2	Unsupported Causality	ATTENUATED	1	PASS	1
UC-3	Unsupported Causality	ATTENUATED	1	PASS	1

Appendix E — Conformance Checklist

Minimal requirements for the demonstrated EVA-HNS behaviour, with status on this PoC. “Met” indicates demonstrated in this controlled run, not certified.

#	Requirement	Status
1	Machine-readable record per decision (Verifiability)	Met
2	Independent post-hoc operation (Transparency, software)	Met
3	Verdict attributable to explicit HNS coordinate	Met
4	Grounding score is an explicit computed metric	Met
5	Token-level evidence cited for each trigger	Met
6	Deterministic and reproducible	Met
7	Correct failure-type assignment on controlled set	Met
8	Immutable hardware substrate (Physical Immutability)	Not met (aspiration)
9	External, blinded, multi-model validation	Not met (future work)
10	Pre-decode prevention via O6	Not met (future work)

Appendix F — Source Note (mapping to the HNS six-part series)

Component	Defined in	Used here as
Five failure types	Volume 1, Appendix A	Targets of the five detectors
HNS-36 layers L1–L6 / categories C1–C6	Volume 2, Ch. 3–4	Layer-jump screen; layer markers
HNS-144 quadrants Q1–Q4	Volume 2, Ch. 7	Category-ambiguity screen
HNS-864 operators O1–O6 (O6 attenuation)	Volume 3	Causal-admissibility screen; actuator
SMS-6 layers / $G(x) / s = \min g_i$	Volume 4, Ch. 2, 10	Grounding margins and composite
EVA three conditions; JSON-LD log	Volume 5	Audit-record schema and verdict logic
Conformance form (SHALL/SHOULD/MAY)	Volume 6	Conformance checklist framing

Accompanying files (reproducibility): `eva_hns.py` (evaluator and scoring functions), `cases.py` (the 15 cases), `run_poc.py` (experiment runner), and the generated artifacts `audit_logs.json`, `audit_logs_jsonld.json`, `results_summary.json`, `before_after.json`, and `results_table.csv`.

Definitional correction logged during preparation. An earlier working reconstruction named the HNS-36 layers as PhysicalOS / CognitiveOS / InteractionOS / EnvironmentOS / LoadOS / PatternOS. That set does not match the source. The authoritative layers, verified against Volume 2, are L1 Physical/Substrate, L2 Cellular/Informational, L3 Cognitive/System 1, L4 Operational/System 2, L5 Micro-Social/Interaction, L6 Macro-Social/Institutional. This PoC uses the authoritative set throughout.